# Series

## Chapter Overview



A series is a sum of a finite or infinite number of terms. In Pure Year 1, $\Sigma$ ("capital sigma") denotes summation.

What we will cover:

1. First $n$ integers: $\sum_{r=1}^{n} r$.
2. First $n$ squares: $\sum_{r=1}^{n} r^2$.
3. First $n$ cubes: $\sum_{r=1}^{n} r^3$.
4. Breaking down complex sums: $\sum_{r=1}^{n}(3 - 2r^2 + 4r)$.
5. Other bounds: $\sum_{r=4}^{2n} r^2$.

We will later prove some results using proof by induction.

**From History of Mathematical Notations, Vol. 2**

438. The sign $\Sigma$ for summation is due to L. Euler[1] (1755), who says, "summam indicabimus signo $\Sigma$." This symbol was used by Lagrange,[2] but otherwise received little attention during the eighteenth century. A widely used notation for the sum of a series is the capital letter $S$; it is given, for instance, in G. S. Klügel's Wörterbuch.[3] The $\Sigma$ to express "sum" occurs in 1829 in Fourier's Theory of Heat,[4] published in 1822, and in C. G. J. Jacobi's[5] elliptic functions of 1829.

Cauchy[6] used three indices $m, n, r$, as in $\sum_{m}^{n} fr$. Alfred Pringsheim[7] marks the sum of an infinite series thus, $\sum_{0}^{\infty} a_r$.

Jahnke[8] adopts in one place, as a substitute for $\Sigma a_\nu$, the simpler sign $\bar{a}$, which he borrows from geodesists, to designate the sum $a_1 + a_2 + a_3 + \cdots$. Additional symbols for sum are given in §410.

## Recap 3

Determine the results by explicitly writing out the terms in each sum.

$\sum_{p=3}^{8} p^2$

$\sum_{r=0}^{5} (7r+1)^2$

## Sum of Ones, Integers, Squares and Cubes 4

These are the four essential formulae for this chapter. The last two appear in the formula booklet, but learn them too.

Sum of $n$ ones: $\sum_{r=1}^{n} 1$

$= n.$

Sum of the first $n$ integers: $\sum_{r=1}^{n} r$

$= \frac{1}{2}n(n+1).$

Sum of the first $n$ squares: $\sum_{r=1}^{n} r^2$

$= \frac{1}{6}n(n+1)(2n+1).$

Sum of the first $n$ cubes: $\sum_{r=1}^{n} r^3$

$= \frac{1}{4}n^2(n+1)^2.$

Notice:

$$\sum_{r=1}^{n} r^3 = \left(\sum_{r=1}^{n} r\right)^2.$$

The sum of the first $n$ cubes equals the square of the sum of the first $n$ integers.

## Quickfire Triangulars! 5

In your head if you can...

$$\sum_{r=1}^{n} r = \frac{1}{2}n(n+1).$$

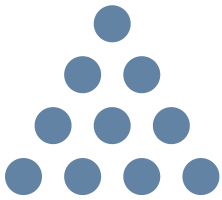
$1 + 2 + 3 + \cdots + 10$

1

$11 + 12 + 13 + \cdots + 20$

$1 + 2 + 3 + \cdots + 99$

$100 + 101 + 102 + \cdots + 200$



## Further Examples of Sums of Natural Numbers 6

Use the formula $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$ to evaluate these textbook examples.

$\sum_{r=1}^{4}(2r-1)$

$\sum_{r=25}^{50} r$

$\sum_{r=1}^{50} r$

Show $\sum_{r=5}^{2N-1} r = 2N^2 - N - 10$, for $N \geq 3$.

Tip: subtract the sum up to one less than the lower limit. Substitute the upper limit, here $2N-1$, for $n$.

## Test Your Understanding So Far 7

Show that

$$\sum_{r=n}^{3n} r = 2n(2n+1).$$

Tip: after writing the subtraction, look for a common factor before expanding.

Proof

## Breaking Up Summations 8

Prove $\sum_{r=1}^{n} kr = k\sum_{r=1}^{n} r$, where $k$ is constant.

Proof

Examples: $\sum_{r=1}^{n} 3r$

$\sum_{r=1}^{n} 4$

## Breaking Up Summations



Prove $\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$.

Proof

Combine this with the constant-multiple property:

$$\sum_{r=1}^{25}(3r + 1)$$

Example

Show $\sum_{r=1}^{n}(3r + 2) = \frac{n}{2}(3n + 7)$.

Further example

Hence evaluate $\sum_{r=20}^{50}(3r + 2)$.

Hence

## Exercise 3A — Core Pure Book 1 Pages 46–47



Exercise 3A — Core Pure Book 1, pages 46–47.

Exercise 3A — Core Pure Book 1, pages 46–47.

Exercise 3A — Core Pure Book 1 Pages 46–47 12

Exercise 3A — Core Pure Book 1, pages 46–47.

Exercise 3A — Core Pure Book 1, pages 46–47.

Exercise 3A — Core Pure Book 1 Pages 46–47 14

Exercise 3A — Core Pure Book 1, pages 46–47.

## Sums of Squares and Cubes 15

(a) Show $\sum_{r=n+1}^{2n} r^2 = \frac{1}{6}n(2n+1)(7n+1)$.

(b) Verify for $n = 1$ and $n = 2$.

Tip: factor out common terms before expanding.

$$\sum_{r=1}^{n} r^2 = \tfrac{1}{6}n(n+1)(2n+1)$$

$$\sum_{r=1}^{n} r^3 = \tfrac{1}{4}n^2(n+1)^2.$$

The polynomial degree of each sum is one higher than that of its terms: squares give a cubic, cubes a quartic.

a

b

## Test Your Understanding 16

Edexcel FP1 (Old), June 2013, Q5

(a) Use standard sums of $r$ and $r^2$ to show

$$\sum_{r=1}^{n}(r+2)(r+3)$$
$$= \tfrac{1}{3}n(n^2+9n+26)$$

for positive integers $n$. (6 marks)

(b) Hence show

$$\sum_{r=n+1}^{3n}(r+2)(r+3)$$
$$= \tfrac{2}{3}n(an^2+bn+c),$$

finding integers $a, b, c$. (4 marks)

a

b

## Harder Exam Question 17

Edexcel FP1 (Old), January 2011, Q5

(a) Use the sums of $r, r^2, r^3$ to prove

$$\sum_{r=1}^{n} r(r+1)(r+5) = \tfrac{1}{4}n(n+1)(n+2)(n+7)$$

for all positive integers $n$. (5 marks)

(b) Hence, or otherwise, find $\sum_{r=20}^{50} r(r+1)(r+5)$. (2 marks)

a

b

## Exercise 3B — Core Pure Book 1 Pages 49–50 18

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Exercise 3B — Core Pure Book 1, pages 49–50.

Exercise 3B — Core Pure Book 1 Pages 49–50 22

Exercise 3B — Core Pure Book 1, pages 49–50.

**Bonus conundrum**

Given that $n$ is even, determine

$1^2 - 2^2 + 3^2 - 4^2 + 5^2 - \cdots - n^2$.

Sum

Alternative